

Integrating Sustainable Design Principles into E-commerce Platforms for Environmental Awareness and Improved motivation in Nigeria

CHAPTER ONE INTRODUCTION

1.1 Background of Study

The formation of the virtual marketplace is a significant change that has widely spread with the development of the internet, which is one critical technological change to almost all sectors across the globe. The digital revolution in the retail of products has led to a substantial change in the practices. Hence, the behavior of businesses and consumers has changed considerably (Nisar & Prabhakar, 2017). A change that was found to have directly emanated from the growth of the internet is the enormous growth in the practice of electronic commerce. This has been the behavior change such that there has been an increase in the number of consumers using virtual markets. Unlike the traditional shopping habits found in nearby stores, it is very much possible for e-commerce to transact globally at any one time since buyers can compare offerings by buyers who are in different cities and countries quite easily. E-commerce fosters product innovations by pushing for information availabilities to competitors and consumers (Nisar & Prabhakar, 2017). This way, the internet is reshaping traditional business paradigms and reinforcing new competitive strategies.

Companies, therefore, have to be extremely careful that they do not disregard environmental conservation in the bid to stretch tentacles of business interests around the globe. To this end, sustainable development prescribes that the needs of humans and processes be weighed against the capacity of the ecosystem to regenerate or reproduce in satisfying such needs for the development of society (Ammenberg and Hjelm, 2013). Unsustainable development occurs when this balance does not exist and consequently creates serious negative impacts on the three pillars of sustainability: environmental, economic, and social—commonly referred to as the "triple bottom line" (TBL).

Increased e-commerce solely reflects the growth of parcel distribution and thus results in more emissions and congestion of the transport system in towns and urban centers. The present share of business-to-consumer (B2C) distribution in e-commerce shipments is close to 56%. This share is bound to upsurge even further as firms enter the online market and demand for parcel distributors increases, auguring increasing emissions and environmental hazards.. The competitive nature of the e-commerce sector pressures logistics service providers (LSPs) to minimize costs, often at the expense of sustainable distribution practices (Arnold et al., 2018).

Companies predominantly focus on the economic dimension, seeking financial growth and revenue. Economic sustainability is frequently viewed as essential for company growth and development. However, sustainable economic growth cannot be achieved if environmental degradation continues or if wealth distribution is inequitable (Bergman & Klefsjö, 2012). Therefore, all dimensional principle of sustainability are equally important. Given the rapid expansion of e-commerce, there is an urgent need for increased awareness and integration of sustainable practices within virtual marketplaces.

1.2 Statement of problem

E-commerce has recorded an accelerated growth rate in Nigeria. The economy is credited with positive implications resulting from the growth of this sector, but there is a green-lighted concern about the environmental impact of this fast expansion. Potential risks include excessive waste, especially packaging, high energy consumption during delivery, and potentially hazardous waste management (Chueamuangphan, Kashyap & Visvanathan, 2020). Beyond the e-commerce industry, the issue also affects broad public understanding of the negative effects of inappropriate trash disposal practices, which is a major obstacle to sustainable waste management in Nigeria. Due to their ignorance of or disinterest in recycling, which has less established environmental benefits, many Nigerians choose to burn their rubbish rather than dispose of it illegally in dump sites. As sustainability measures have not been implemented, these practices can actually lead to air pollution and have an impact on climate change, soil contamination, hazards to public health, and the depletion of resource bases.

A sustainable society is the custodian of the environment and responsible in the social sector. Sustainable development connotes a balanced approach involving the human system and the natural system. This allows Nigerian e-commerce brands to make their platform integrated with the principles of sustainable design for example, promoting environmental awareness, motivating customers to adopt eco-friendly choices and sustainable practices, and using the medium to save on production costs, operational efficiency, and retaining customers. E-commerce companies are able to accomplish this by lowering the amount of resources and materials they need to reproduce, which can shorten lead times and simplify processes. Reducing delivery times could potentially result in happy customers and a decrease in returns.

The present dissertation centers on the potential of e-commerce platforms to act as catalysts for positive change, encouraging more conscientious and sustainable consumption, since sustainable design can assist in considering environmental concerns at every stage of a product's life cycle, from production to disposal. These sustainable design principles, which can lessen environmental footprints and contribute to consumer education and public awareness, help e-commerce platforms minimize their ecological footprint.

1.3 Objective of the Study

The primary aim of the research is to study how sustainable design principles in the e-commerce sector can increase awareness and, therefore, interest by consumers in sustainable practices in Nigeria. In this regard, the study shall:

1. Evaluate the extent of awareness by Nigerian e-commerce users on sustainable practices, primarily focusing on ways of disposing waste.
2. Identify those barriers that have mitigated awareness and the diffusion of sustainability practices in the e-commerce industry in Nigeria.
3. Develop and propose sustainable design improvement features of the e-commerce platforms to generate an adequate level of awareness and promotion around sustainable practices to persuade customer engagement and their buying behavior.

1.4 Research questions

1. How aware are Nigerian e-commerce consumers of practices that support sustainability, in particular, methods of disposal of used goods?

2. What hitches influence and limit the awareness of sustainability practices in the Nigerian e-commerce sector?
3. What sustainable design strategies can e-commerce players use to communicate and promote better sustainable practices that drive consumers' participation in Nigeria?

1.5 Aim and Scope of the Study

This dissertation argues how sustainable design implementations within the Nigerian e-commerce industry will create awareness among users and convince them to engage in such sustainable design acts. It focuses on how implementing sustainable design features in the systems is likely to orient and influence users to become aware of and engage in environmental challenges.

1.6 Dissertation structure

The paper is divided into six chapters, each tailored to its specific purpose:

Chapter 2: Literature Review - This chapter will focus on reviewing and evaluating previous works by other scholars and relevant subjects to identify theories, ideas, and gaps related to this topic.

Chapter 3: Methodology - This chapter outlines the research strategy, approach, and methods employed in this study, encompassing both quantitative and qualitative methods.

Chapter 4: Results and Analysis - This chapter presents the findings from the research methods, including survey data and interviews, and provides detailed information based on the analyzed data.

Chapter 5: Discussion - In this chapter, the findings are discussed and compared with the literature review.

Chapter 6: Conclusion - This chapter provides a broad summary of the study's findings, addressing how the research questions and problems were resolved, highlighting the overall contributions of the study, and offering recommendations for future research.

Chapter two

Literature review

2.0 Introduction

This literature review centers on two parts: E-Commerce, first of all be introduced with its definition and types, and secondly, with this foundation constructed, the relationship of E-commerce platforms in Nigeria, their evolution, development, current state, and the prospects and challenges of E-Commerce within/by Nigeria. In the second part, the relationship/or usage in e-commerce and the current state of sustainable design principles are addressed in explaining the concept between sustainability and the e-commerce trajectory in Nigeria.

2.1 Concept of E-Commerce

This section will lay the groundwork for this literature study by providing an overview of the definition, categories, and fundamental ideas of e-commerce.

2.1.1 Definition of E-Commerce

E-commerce, as defined by the Organization for Economic Cooperation and Development (OECD, 2009), is the buying and selling of products or services through computer networks using order placing or reception mechanisms that are particularly designed for this purpose. Even though these techniques are used to initiate transactions, online payment and subsequent delivery of products or services are not a must these interactions might take place between private or public organizations, residences, corporations, governments, and individuals (OECD, 2011). Despite adding price comparison and information exchange as crucial components of the E-Commerce ecosystem, Khan (2016)'s definition of E-Commerce still closely aligns with the OECD's definition.

2.1.2 Types of E-Commerce

E-commerce can be classified according to the parties that are engaged in the transaction. Schneider (2016) delineates five discrete groups, which include "business-to-business, consumer-to-consumer, business-to-government, transactions and business processes, and business-to-business". These two most well-known types of sales are business-to-business (B2B) and business-to-consumer (B2C), with B2B making up a significantly higher share of sales than B2C. E-commerce players can also move across other categories. For example, intermediary items can be traded via business-to-business (B2B) transactions before going through B2C channels to reach

end users. These key areas will be covered in detail, together with an examination of the dynamics of consumer-to-consumer (C2C) technology.

B2B; According to Schneider (2016), business-to-business (B2B) commerce involves organizations selling goods or services to other businesses. A quintessential example is Microsoft supplying software to Volkswagen, constituting a B2B transaction. According to Mourya & Gupta (2015), this mode of E-Commerce contributes to over 90% of the total E-Commerce volume.

B2C; B2C commerce is the practice of businesses selling products or services directly to consumers (Schneider, 2016). According to dos Santos et al. (2017), B2C transactions outnumber other E-Commerce categories in terms of frequency, and this situation occurs when a customer buys anything from websites such as Amazon (Mourya & Gupta, 2015).

C2C; C2C commerce includes people who trade goods and services online, while being the smallest category in terms of sales. One example is the German digital marketplace "eBay Kleinanzeigen," which allows for peer-to-peer transactions. Notably, C2C transactions frequently cross over into the B2C sphere since people acting as sellers may be interpreted as businesses (Schneider, 2016).

2.2 E-Commerce platform in Nigeria

This section provides an overview of the current state of e-commerce in Nigeria, beginning with its development and current condition, followed by a discussion of the challenges faced.

2.2.1 Development of E-Commerce in Nigeria

2.2.1.1 Evolution of E-Commerce in Nigeria

The growth of e-commerce has taken unique paths across various regions globally, contrasting sharply with the rapid expansion observed in developed countries since 1995. In Nigeria, the adoption of this digital revolution was initially slow, as evidenced by the fact that in 1999, there were only 200,000 internet users, representing a mere 0.16% of the population (The Ministry Of Communication Technology, 2012; World Data, 2022). To address this lag, the Nigerian government implemented strategic initiatives, including the introduction of the National Technology Policy in 2000. According to the Federal Republic of Nigeria (2001), this policy aimed

to establish Nigeria as a hub of information technology expertise in Africa by 2005, leveraging IT to foster sustainable development and enhance international competitiveness.

Through initiatives like the State Information Infrastructure (SII), the policy ensured that IT innovations were implemented to enhance global competitiveness, foster wealth creation, alleviate poverty, create jobs, and improve education. These objectives were further advanced in 2007 with the establishment of the National Information Technology Development Agency (NITDA), which took steps to create national internet usage policies and guidelines (NITDA, 2007). The Ministry of Communication Technology (2012) reported that, as a result, internet usage surged to 23.48% by 2010. This increase prompted the government to develop the National Information Communication Technology (ICT) strategy in 2012, aimed at propelling Nigeria into the top 20 global economies.

However, in comparison to its peers, Nigeria's digital landscape remained underdeveloped despite notable gains. Ojiako et al. (2012) found that the adoption of internet services remained low, and the Nigeria Vision 2020 highlighted the early stages of digitization, including e-commerce (National Planning Commission, 2009). This position is supported by data from the United Nations Conference on Trade and Development (UNCTAD), which states that by 2013, e-commerce retail sales made up a very small portion of all retail sales. Nigeria, however, only ranked 101 on the 2014 UNCTAD B2C E-Commerce Index (UNCTAD, 2015).

2.2.1.2 Current status of E-Commerce in Nigeria

The electronic commerce (e-commerce) environment in Nigeria is starting to show the results of years of modifications. The country's internet user base is steadily increasing as a result of ongoing expenditures in the Information and Communication Technology (ICT) industry. Notably, internet connectivity in Nigeria is gradually improving with the construction of broadband infrastructure. Furthermore, the public is becoming more accustomed to and using e-commerce platforms for private transactions as a result of the government's growing acceptance of e-governance services. This tendency has a major role in advancing the e-commerce industry and revealing substantial long-term economic benefits. Early signs of improvement are already apparent, as seen by the notable rise in popularity of online markets like Konga and Jumia in recent years (Ibam et al., 2018). Nigeria has consequently become known as a hub of e-commerce innovation within the African continent, currently leading the way (Igwe et al., 2020).

However, Nigeria's e-commerce industry continues to be marked by constant change. There is still a sizable proportion of younger demographics among e-commerce users. Thus, in 2020, the market was centered mostly on the following industries: hotels and airlines (22%), electrical goods (17%), and health and beauty (23%). According to Sasu (2022b), this suggests that luxury goods are the main focus of e-commerce platforms, whereas ordinary necessities are still primarily traded offline. Nigeria has an approximately 50% internet penetration rate and a roughly 26% online purchasing percentage of the country's overall population as of 2020 (EcommerceDB, 2021). (Sasu, 2021). As a result, slightly more than half of people who have access to the internet make purchases online.

The growing importance of e-commerce companies in the country's online marketplace is reflective of the rising e-commerce sector in Nigeria. For instance, Kemp (2022) reports that Konga ranks 10th, with 4.8 million unique visitors, and Jumia ranks 18th, with 1.95 million unique visitors, among the top 20 most visited websites in Nigeria. That shows a good level of user engagement with e-commerce websites that are important to meet their needs. Moreover, the academic interest in Nigeria's e-commerce sector is duly represented by Ejemeyovwi et al. (2019), Okolie & Ojomo (2020), Sasu (2022), Ayo et al. (2011)

2.2.1.3 Prospects for E-Commerce in Nigeria

Nigeria is strategically positioned to capitalize on significant opportunities for e-commerce growth, driven by the rising number of internet users and subsequent increase in e-commerce participants. Much of this potential remains untapped, suggesting that Nigeria has even greater capacity for e-commerce development compared to more developed nations. Nnamdi Ekeh, Co-CEO of Konga, foresees rapid growth in African e-commerce, stating, "eCommerce in Africa is set to take off to unprecedented heights in the next decades" (Forbes Africa, 2022). Projections indicate that Nigeria's internet penetration will increase by over 1.5% annually, reaching nearly 60% by 2026 (Johnson, 2021).

Moreover, the Nigerian government has ambitious targets, aiming to boost e-commerce spending from \$12 billion in 2021 to \$75 billion by 2025 (International Trade Administration, 2021b). Research suggests that countries with higher internet penetration and advanced technological infrastructure experience faster e-commerce growth, positioning Nigeria ahead of other African nations (Zatonatska et al., 2019).

Nigeria has demonstrated its leadership in digital currency with the creation of the E-Naira, spearheading the "cashless economic transformation" in Africa (Forbes Africa, 2022). The National Intelligence Council (2021) has confirmed that Nigeria's leadership in e-commerce in the area is expected to intensify its impact on the continent's economy. Furthermore, this effect promotes economic cooperation and advances adjacent nations.

2.2.2 Challenges of E-Commerce platform in Nigeria

There are requirements that must be fulfilled in order to develop a profitable e-commerce business. These requirements might be challenging to meet, particularly in developing nations. Incorporating E-Commerce in developing nations presents unique obstacles since their societies have developed differently from those of industrialized nations.

Table 1: E-Commerce Challenges in Nigeria

Challenge in Nigeria	Literature
1 Deficits in Technological Infrastructure	(Ajao et al., 2018; Rabiou et al., 2019; Khan & Uwemi, 2018)
2 Inadequate Internet Infrastructure	(Gillwald et al., 2018; Khan & Uwemi, 2018; Schneider, 2016; Rabiou et al.)
3 High Implementation Costs	(Choshin & Ghaffari, 2017; Lawal & Ogbu, 2015; Ayo et al., 2011; Gillwald et al., 2018; Nejadirani et al., 2011)
4 Insecure Payment Systems	(Ayo et al., 2011; Khan & Uwemi, 2018; Rabiou et al., 2019)
5 Underdeveloped Logistics	(Lawrence & Tar, 2010)
6 Concerns Regarding Security	(Ibami et al., 2018; Igwe et al., 2020; Khan & Uwemi, 2018; Ayo et al., 2011)
7 Privacy Deficits	(Khan & Uwemi, 2018; Lawal & Ogbu, 2015)
8 Customer Satisfaction Challenges	(Choshin & Ghaffari, 2017)
9 Lack of Website Information	(Ayo et al., 2011; Igwe et al., 2020)
10 Limited Awareness and Knowledge	(Choshin & Ghaffari, 2017; Lawal & Ogbu, 2015; Rabiou et al., 2019)
11 Issues with Trust and Confidence	(Ajao et al., 2018; Igwe et al., 2020; Rabiou et al., 2019)
12 Shortage of Trained Personnel	(Ajao et al., 2018)

There are some very difficult challenges facing business entities looking to get into Nigeria's e-commerce market. Rabiou et al. (2019) draw attention to the obvious difficulties in obtaining the necessary technology infrastructure in developing countries such as Nigeria, mainly because of the pervasive poverty that exists in the country. As such, a sizeable portion of the population does not have the financial resources to purchase internet-enabled devices. In addition, the lack of a dependable, national internet connectivity infrastructure is a major barrier to E-Commerce initiatives (Gillwald et al. 2018),

Gillwald et al. (2018) emphasize the additional financial burden associated with buying ICT equipment for E-Commerce activity. The burden is exacerbated by import duties, which are set by prices on the global market and are often unaffordable for a large number of Nigerians. FinTech shortcomings make Nigeria's transactional environment even more challenging. This is especially true of online payment mechanisms, which are still far from ideal and have significant shortcomings. Additionally, rural areas' logistical difficulties make it more difficult to get prepared for e-commerce, which complicates efforts to encourage client uptake.

According to Khan & Uwemi (2018), security issues are a major issue in the e-commerce sector. They stress how important it is to allay customers' worries about privacy and security in order to build user confidence. The burgeoning e-commerce sector in Nigeria continues to struggle with low customer satisfaction, which is exacerbated by widespread issues including badly designed and functioning websites. As a result, the general public still views traditional trading channels favorably rather than accepting e-commerce as the main form of business.

Rabiou et al. (2019) emphasize the importance of awareness, especially in rural areas where there is still a lack of familiarity with E-Commerce principles, which hinders the general use of this technology. Furthermore, when customers carefully consider whether to accept these services, a history of mistrust towards businesses and the slow development of online payment methods impede market penetration. The operational issues within the E-Commerce business are exacerbated by the lack of skilled staff who can manage these platforms (Ajao et al., 2018).

2.3 Sustainable Development

Businesses must carefully consider the sustainability of their actions in order to protect the integrity of the world while they pursue global outreach and business operations expansion (Hariram et al., 2023). The foundation of sustainable development is striking a fine balance between the requirements of humans and ecosystems' ability to recover and thrive, while simultaneously promoting social progress. Should each of the three principles of sustainability not be balanced, development may become unsustainable and lead to a variety of negative consequences under emerging economic paradigms (Hariram et al., 2023).

The concept of sustainability is developing as an emerging important dimension in e-commerce because it recognizes that e-commerce companies need to adopt sustainable design principles to cope with problems in the economic, social, and environmental domains (Oláh et al., 2018). This entails applying sustainable design principles to create products, services, and systems that minimize environmental impact while promoting corporate social responsibility throughout their entire life cycle. One key concept is Life Cycle Assessment (LCA), which demonstrates the environmental impact of products from manufacturing to disposal. LCA also assists businesses in finding ways to utilize less resources, cut down on waste and pollution, and reduce pollutants.

2.3.1 Principles of Sustainable Design

The fundamental design principles of sustainability are typically divided into three categories: social, economic, and environmental (Purvis et al., 2019). However, the interaction and synergy between these ideas, made possible by global cooperation, rather than their simple definition, hold the key to a more promising future. Hart's wise words,

"Those who think sustainability is only a matter of pollution control are missing the bigger picture," have a deep resonance.

Therefore, each country must build a coherent political agenda that addresses all three aspects of sustainability development and is specific to the needs of its own setting (Purvis et al., 2019).

2.3.1.1 Environmental Principle

The three principle design of sustainability are interconnected, and each is essential to promoting a more sustainable global ecosystem (Hariram et al., 2023). The environmental principle is

fundamental to this approach as it defines the basic conditions for all life on Earth and sets the basic boundaries for the other principle. According to Purvis et al., (2019), this Principles focused on the outside forces that affect biodiversity, natural resource management, and the environment. Thus, promoting a circular economy, environmental sustainability centers on the development of production techniques and products that reduce environmental effect throughout the course of their lifecycle (Purvis et al., 2019).

2.3.1.2 Economic Principle

According to Hariram et al., (2023), the economic principle design comprises effective use of resources, promotion of stability, and creation of long-lasting economic frameworks. A sustainable approach demands attention to long-term economic sustainability even though many businesses target short-term advantages to appease customers. Consumer demand has a significant impact on the economy, thus business enterprises are compelled to embrace more environmentally friendly and sustainable methods in order to gain a competitive edge. This is due to consumers' growing preference for items with ethical designations. According to (Purvis et al., 2019). Satisfied customers, a strong international market free of trade obstacles, and prudent use of natural resources and systems are the means by which economic sustainability is thereby promoted.

2.3.1.3 Social Principle

Although there isn't a widely accepted paradigm for the social dimension, it includes aspects of human needs and cultural growth and is emphasized by a quest of equity (Purvis et al., 2019). Fundamentally, access to necessities including housing, drinkable water, work, and healthcare must be available to all people. Societies with these necessities and strong social capital are better able to implement transformative changes away from harmful practices, including the patterns of consumption that are common in today's world (Hariram et al., 2023; Purvis et al., 2019).

2.3.2 Integrating Sustainable Design into E-Commerce

The integration of sustainable design concepts in e-commerce can yield several advantages, such as decreased operational expenses, risk mitigation, and enhanced brand perception upon market entry. At every point of the product life cycle, including conceptualization, marketing, and distribution, businesses can apply sustainable design principles. Furthermore, companies can meet consumer demands for ethical and environmentally friendly goods and services while lessening

their impact on the environment by implementing eco-design, green packaging, energy-efficient manufacturing, sustainable sourcing, and circular economy models (Ghisellini et al., 2016). Even with all of these advantages, technological limitations, market dynamics, and consumer behavior continue to be the main obstacles to sustainable design. Stakeholder collaboration, investment in research and development, and policy support are key to overcoming these barriers. Opportunities for businesses lie in increased concerns regarding sustainability issues, the status of green technologies, and shifting consumer preferences.

2.4 Consumer Knowledge

The fact that consumers are becoming more conscious of the environment and concerned about it as a result of realizing the influence of their purchase decisions is another element that businesses must consider (Gajdzik, Jaciow, & Wolny 2023). A growing number of consumers are learning to take environmental issues into account when they shop, such as by asking if a product's packaging is made of recycled materials, purchasing more environmentally friendly products, and generally demonstrating a willingness to pay more for an environmentally friendly product. According to Kianpour et al. (2013), "green" or environmentally friendly products refer to any type of product, service, or policy that has no negative impact on the environment or only minor negative effects.

There has also been a discussion of how useful educational tools such as Eco-labels motivate customers to be more sustainable. With their easily comprehensible symbols or badges that indicate a product's level of sustainability such as its ability to be recycled, its energy efficiency, and the sustainability of the materials used eco-labels help consumers make the transition to a circular economy. Meis-Harris et al. (2021) added that eco-labels' instantly identifiable badges or symbols that indicate a product's recyclable status or level of sustainability may also function as cues. These labels can be applied by online retailers to their items to indicate that they are environmentally friendly; this may create awareness and instill motivation in customers toward making smart purchases (Hussain et al., 2020).

According to Hayat, Hussain, and Lohano (2020), obtaining Eco-label certifications through the development of trust and honesty can help firms stand out from the competition, draw in environmentally conscious clients, and foster brand loyalty. Because products must adhere to tight standards and specifications throughout their entire life, this could further promote supply chain

transparency. This drive to continuously generate novel concepts aids in the creation of products and processes for a more ecologically sustainable circular economy.

Businesses are also making an effort to take a more ecologically friendly stance in response to growing concerns. For example, businesses are making more ecologically friendly products, suppliers are using fewer plastic bags and other non-green packaging materials, and businesses are allocating funds to continue improving their operations (Kianpour et al., 2013). It is crucial for businesses to alter customer attitudes by informing, communicating, and educating customers about the benefits of purchasing environmentally friendly products. The goal is not simply to draw in customers and raise awareness, but also to show them that individual acts matter and that corporations are not the only ones who have to safeguard the environment.

2.5 E-commerce and Sustainability

E-commerce is growing at a rapid pace, which offers both opportunities and challenges for businesses looking to operate sustainably. According to Macchion et al. (2017), compliance with sustainable development concepts and legislation is becoming more and more important as consumer and society awareness rises. Ethical advancement demands that the entire supply chain be taken into account and that the economic, environmental, and social principle design of sustainability be integrated (Macchion et al., 2017; Mangiaracina et al., 2015). Finding a balance between people, the environment, and profits can lead to more sustainable results, which will improve a business's performance as a whole (Shao & Liu, 2012). Zhu et al. (2012) state that combining e-commerce with sustainability efforts is complicated by the need to navigate a variety of international rules.

2.5.1 E-commerce and Environmental principle design Impact

Debates persist regarding e-commerce's impact on environmental sustainability, particularly concerning transportation-related emissions (Carrillo et al., 2014; McLeod et al., 2006). The surge in online shopping has increased shipment volumes, predominantly in van traffic, albeit with mixed environmental implications (Mangiaracina et al., 2015). Home deliveries have shown promise in reducing individual trips and emissions, countering some negative environmental effects (Carrillo et al., 2014). However, fragmented deliveries resulting from orders across multiple channels contribute to inefficiencies and increased transportation emissions.

(Mangiaracina et al., 2015). The global accessibility facilitated by e-commerce amplifies transportation routes and emissions, exacerbating environmental concerns (Nisar & Prabhakar, 2017).

Moreover, the proliferation of parcel distribution, driven by e-commerce growth, adversely impacts urban quality of life due to heightened emissions and congestion (Arnold et al., 2018). Pressures to minimize transportation costs and delivery times exacerbate sustainability challenges, particularly in the last-mile delivery segment (Arnold et al., 2018). Collaborative efforts among logistic service providers offer potential solutions to optimize last-mile delivery efficiency and reduce emissions (Arnold et al., 2018). Transitioning to alternative vehicles, such as electric and hybrid models, presents an opportunity to mitigate CO₂ emissions in e-commerce logistics (Mangiaracina et al., 2015). Initiatives like Sainsbury's adoption of electric delivery vans and Chinese companies utilizing electric bicycles demonstrate promising strides toward greener logistics (Mangiaracina et al., 2015). The packing tactics used in e-commerce aggravate environmental issues by using a lot of resources for each order and increasing trash. Since many shipments are required for a single order, fragmented supply points exacerbate environmental impacts even further (van Loon et al., 2015).

2.5.2 E-commerce and Economic Sustainability

E-commerce offers compelling economic benefits, including cost reduction, profit increase, operational efficiency, and faster time to market (Mangiaracina et al., 2015; Theyel & Hofmann, 2015; Deljavan Anvari & Norouzi, 2016). However, the competitive nature of online marketplaces can lead to initial financial losses for e-commerce firms (Yang et al., 2016). Continuous innovation is essential to meet consumer demands and maintain competitiveness (Macchion et al., 2017). Building consumer trust and loyalty is crucial in a crowded marketplace where brand familiarity influences purchasing decisions (Sullivan & Kim, 2018).

Despite challenges, e-commerce has demonstrated positive economic impacts at the enterprise, sector, and national levels, driving economic growth, labor productivity, and consumer welfare (Deljavan Anvari & Norouzi, 2016).

2.5.3 E-commerce and Social Sustainability

E-commerce facilitates global communication and knowledge transfer, expanding market access and fostering social innovation (Cui et al., 2017). Social innovations enabled by e-commerce address various societal needs, including poverty alleviation and cultural preservation (Pol & Ville, 2009; Phills et al., 2008). For example, e-commerce platforms empower rural villagers to sell their products globally, contributing to local economic development and poverty reduction (Cui et al., 2017). While e-commerce bridges gaps between developed and developing countries, it also exacerbates economic disparities and job displacement concerns (Lozano, 2007; Autor, 2013). However, it also creates opportunities for digital entrepreneurs and individuals with digital literacy (Biagi & Falk, 2017). Balancing these social implications remains a key challenge for sustainable e-commerce development.

2.5.4 The Role of E-Commerce Platforms In Promoting Sustainability

E-commerce platforms can significantly contribute to solving environmental problems by encouraging their customers to adopt eco-friendly waste disposal methods. Their ability to connect with vast numbers of people online makes them ideal for spreading awareness about recycling and sustainable consumption. Utilizing their digital infrastructure, these platforms can educate customers on specific eco-friendly actions and motivate them to engage in practices that benefit the environment (Oláh et al., 2018).

E-commerce platforms have the potential to play a significant role in promoting recycling programs and facilitating product returns for certain types of products. Research indicates that setting up take-back and return programs, especially for used technology, faulty and unused electronics, or printer cartridges, can encourage consumers to properly dispose of their waste products and aid in resource recovery (Wang, Zhu, Yao, & Sun, 2020). This initiative can motivate customers to return used items to the platform for recycling or refurbishment. For Nigerian e-commerce sites, partnering with established recycling businesses and informal trash collectors, who are pivotal in waste management in Nigeria, could be a promising approach. However, it is important to recognize that not all e-commerce businesses in Nigeria may afford these options, making such initiatives seem exclusive and potentially unfair to smaller business owners.

The table Below is showcasing e-commerce platforms from developed countries that have successfully implemented sustainable practices and effectively communicated these efforts to their consumers.

Table 2 Sustainability initiatives of e-commerce brands.

PLATFORM	SUSTAINABLE DESIGN EFFORTS
AMAZON	Amazon has committed to reaching net-zero carbon emissions by 2040. This ambitious goal includes the Climate Pledge Friendly program, which promotes sustainable consumer practices. Key investments in solar energy and electric delivery vehicles highlight Amazon's efforts to create a more sustainable value chain from production to delivery.
ALIBABA	For the reduction of carbon emissions and to aid green supply lines, Alibaba suggested starting programs such as the Green Leadership Program and Alibaba Cloud's Carbon Management Solution, in addition to efforts put into eco-friendly packaging and logistics improvement.
EBAY	eBay unveiled its Retail Revival program in Wolverhampton in 2018, which supported many small and medium-sized businesses that were selling sustainable goods. This Retail Revival program has allowed small and medium-sized companies worldwide to tap the global marketplace of eBay, drawing new customers and engaging them in sustainable practices
ETSY	Etsy plants sufficient trees and invests in green energy projects enough to act as a carbon neutralizer for all the emissions made during shipping. Encouraging clients to know about these stages towards attaining sustainability is essential in raising their understanding of the advantages of eco-friendly practices and sustainability, which in turn may inspire them. Customers are encouraged to utilize eco-friendly packaging by Etsy.

Source: Salest blog 2023

2.6. Conceptual link-chain between sustainability and e-commerce Trajectory in Nigeria

Dai et al. (2018), Deljavan et al. (2016), Macchion et al. (2017), and Olah et al. (2016) have contributed significantly to the understanding of the various environmental impacts associated with the integration of e-commerce into business operations. Their research has highlighted several

key factors such as increased trade volume, extensive use of transportation infrastructure, waste management challenges, initial investment requirements for establishing e-commerce platforms, the necessity of a robust legal framework, and the risks of data and identity theft and other forms of cybercrime. These factors collectively represent the complex interplay between economic, social, and environmental considerations in the context of e-commerce sustainability.

From an economic perspective, e-commerce offers organizations the opportunity to expand their customer base globally, potentially leading to increased revenue and profitability. Additionally, it promotes the use of electronic devices and can result in cost savings for both businesses and consumers. Moreover, e-commerce facilitates social interactions, particularly for businesses engaged in social-related activities. For instance, the rise of online entertainment platforms enables artists to reach a wider audience and sell their products more efficiently. Furthermore, e-commerce contributes to environmental sustainability by reducing the need for physical travel by consumers.

Notwithstanding the potential benefits of e-commerce, a number of obstacles prevent it from being widely adopted, especially in areas with inadequate internet connectivity. For example, in Nigeria, internet penetration was only 41.8 percent in 2019 (Clement, 2020), representing significant progress compared to previous years but still below the global average. Forecasts indicate that by 2025, the percentage of Nigerians who use the internet is predicted to reach 65.3%, demonstrating a trend in the direction of increased accessibility and communication. Nonetheless, challenges remain regarding the equitable distribution of internet access and the mitigation of associated risks such as cybersecurity threats.

To ensure the sustainable integration of internet resources into business operations, comprehensive financial inclusion across primary, secondary, and tertiary levels is imperative (Okunlola et al., 2020b). Historically, there has been a deficiency in the adoption and utilization of various e-commerce platforms among business proprietors. Notably, platforms such as mobile money (M-money), mobile agents (M-Agents)/mobile cash (Mcash), E-Billspay, Automatic Teller Machine (ATM) transfers, point-of-sale (POS) systems, web-pay, Remmita, Interswitch, NAPS, NIP, Konga, Jumia, Jiji, Olx, Central-Pay, Opay, Safeborder, Uber, Taxify, among others, initially garnered limited acceptance. However, their impact has gradually extended across domains such as education, marketing, transportation, advertising, commerce, and governance. Recently, there has been a growing recognition of the potential for these e-commerce platforms to sustain businesses,

albeit primarily by adding value (Ogbo et al., 2019; Ibam et al., 2018). Similar trends in platform utilization have been observed across various sectors.

Table 3. Some e-commerce channels in Nigeria

Uber	Ajebo Market	M-Agent	Remitta	Olx
Next cash & Carry	Webpay	Konga	Jiji	Jumia
Taxify	Opay	Safe Brder	E-bills pay	

Table 3 show a selection of e-commerce platforms and channels utilized for business transactions in Nigeria. While this list is not exhaustive, it encompasses commonly utilized platforms. Notably, these channels cater to various sectors in Nigeria, including transportation, consumables, financial transactions, among others. Overall, a comprehensive review of existing literature regarding the sustainability of businesses through e-commerce applications in Nigeria yields mixed conclusions, debunking some myths while affirming others.

2.7 Research Gaps and Need for Further Research

More investigation is required to determine how effectively various sustainable design concepts might function in the Nigerian e-commerce environment. The fact that most Nigerians are not taught much about sustainability and do not know much about it themselves is a major issue. People's perceptions of the benefits and simplicity of adopting sustainable behaviors may shift as a result of additional study and application. Researchers could investigate in-depth how design plans can incorporate cultural perspectives on sustainability. Studies of this kind might also examine the ways in which various public awareness campaigns and reward schemes could eventually drive and alter consumers' behavior. It is imperative to examine the impact of e-commerce platforms on the Nigerian environment, taking into account factors such as the reduction of packaging waste and the carbon footprint associated with delivery. It is necessary to conduct research on the creation of scalable and culturally appropriate recycling programs.

In conclusion, e-commerce can assist reduce the harm it causes to the environment in underdeveloped nations like Nigeria by using sustainable design principles. E-commerce platforms play a critical role in customer education, waste management promotion, and

establishing relationships with environmentally conscious or recycling businesses. E-commerce platforms in poor nations, such as Nigeria, may empower users to make sustainable decisions by fusing innovation in design, education, and collaboration.

Chapter Three

Research Methodology

3.1 Introduction

This chapter outlines the methodology employed to address the research questions and elucidates the data collection methods utilized. The selected approach endeavors to explore and comprehend how the incorporation of sustainable design into e-commerce platforms can bolster environmental consciousness and encourage sustainable consumer behavior in Nigeria.. This chapter provides an in-depth explanation of the selected study methods, data collection and analysis procedures, and ethical considerations encountered during the research process. A mixed-methods approach was adopted, combining quantitative and qualitative methods to comprehensively understand e-commerce consumer behaviors, the influence of design elements, and strategies to foster consumer engagement in sustainable practices.

3.2 Research Design

A mixed-methods research design was employed, integrating both quantitative and qualitative methods to capture perspectives and measure current awareness levels regarding sustainability practices. This approach facilitates a multifaceted examination of consumer behavior and the impact of sustainable design, enabling data from one method to complement and enhance the findings from the other.

3.2.1 Quantitative Methods

A quantitative survey was conducted to assess the knowledge, attitudes, and practices of Nigerian online shoppers concerning sustainable e-commerce practices. The survey aimed to gauge awareness levels on sustainability topics. A structured questionnaire, designed using Microsoft

Office Forms, was distributed electronically. The questionnaire included a mix of open and closed-ended questions based on established principles.

3.2.2 Qualitative Methods

In addition to the survey, qualitative interviews were conducted with Nigerian e-commerce users to explore their experiences, motivations, and challenges related to sustainable consumption. Structured questions guided these interviews, focusing on participants' experiences, decision-making processes, and perceptions of sustainable choices in online shopping. The interviews were conducted in person, recorded with participants' consent, and subsequently transcribed for analysis.

3.3 Sampling Strategy

3.3.1 Quantitative Survey

A non-probability sampling technique, specifically convenience sampling, was used to select participants for the quantitative survey. This method involved gathering data from easily accessible individuals who could participate in the online survey. To ensure representative results, participants from various regions of Nigeria and diverse social backgrounds were included.

3.3.2 Qualitative Survey

Purposeful sampling was employed to select participants for the semi-structured interviews. This approach ensured the selection of individuals with relevant experiences to provide rich and insightful responses. Participants were chosen based on their frequency of online shopping and their interest in environmentally friendly practices.

3.4 Data Collection Methods

- **Quantitative Data Collection:** A structured questionnaire, distributed electronically via Microsoft Office Forms, was used to gather quantitative data from Nigerian online shoppers. The questionnaire comprised both open and closed-ended questions to capture a comprehensive understanding of participants' awareness, attitudes, and behaviors regarding sustainable e-commerce practices.

- **Qualitative Data Collection:** Semi-structured interviews were conducted with a subset of Nigerian e-commerce users. These interviews explored participants' experiences, motivations, and challenges related to sustainable consumption. The interviews were recorded with participants' consent and transcribed for thematic analysis.

3.5 Data Collection Instruments

- **Questionnaires:** The structured questionnaire used in the quantitative survey was developed based on established principles and included a mix of open and closed-ended questions to capture detailed information on participants' knowledge, attitudes, and practices.
- **Interview Guides:** The semi-structured interview guide used in qualitative interviews was designed to elicit rich, detailed responses from participants about their experiences and motivations related to sustainable consumption in e-commerce.

3.6 Data Analysis Techniques

3.6.1 Quantitative Data

The quantitative data will be analyzed using Microsoft Excel or spss, which facilitated automatic categorization and grouping of responses from Microsoft Forms. The data were analyzed using thematic analysis to identify key trends and patterns in customer awareness and behavior. The results were presented using numerical and percentage-based summaries to highlight the main trends and characteristics of the sample group.

3.6.2 Qualitative Data

The qualitative data from interview transcripts will be transcribed and analyzed using thematic analysis. This analysis aimed to assess participants' awareness levels, identify motivating factors, and explore the features that could encourage sustainable behavior when integrating sustainable design elements into e-commerce platforms.

3.7 Validity and Reliability

To ensure the validity and reliability of the research, several measures were implemented. The survey and interview questions were pre-tested to refine the instruments and ensure they

effectively captured the desired information. Additionally, the mixed-methods approach allowed for cross-validation of data, enhancing the robustness of the findings.

3.8 Ethical Considerations

Ethical considerations were paramount throughout the research process, particularly during interviews and the distribution of questionnaires. The following ethical guidelines were adhered to:

3.8.1 Informed Consent

All participants were provided with clear information about the study's objectives and the nature of their involvement. Participants were informed that their perceptions and ideas were the primary focus. Written consent was obtained from interview participants after briefing them about the study.

3.8.2 Security

No personal information was collected from interview participants, and responses to the quantitative survey were gathered anonymously. Participants were assured that their information would be kept confidential during the research and reporting stages, with data anonymized to prevent identification.

3.8.3 Right to Withdraw

Participants were informed of their right to withdraw from the study at any time without any repercussions. They were assured that their data would be deleted upon withdrawal if they chose to discontinue participation. By adhering to these ethical guidelines, this research aims to protect participant privacy throughout the study.

REFERENCE

- Ayo, C. K., Adewoye, J. O., & Oni, A. A. (2011). Business-to-consumer e-commerce in Nigeria: Prospects and challenges. *African journal of business management.*, 5(13), 5109-5117.
- Carrillo, J.E., Vakharia, A.J., & Wang, R. (2014). Environmental implications for online retailing. *European Journal of Operational Research*, 239, pp. 744-755.
- Choshin, M., & Ghaffari, A. (2017). An investigation of the impact of effective factors on the success of e-commerce in small- and medium-sized companies. *Computers in human behavior*, 66, 67-74. <https://doi.org/10.1016/j.chb.2016.09.026>.
- Clement, J. (2020). *Worldwide digital population as of April 2020*. <https://www.statista.com/statistics/617136/digital-populationworldwide>
- Dai, Y. N., Viken, G., Joo, E., & Bente, G. (2018). Risk assessment in e-commerce: How sellers' photos, reputation scores, and the stake of a transaction influence buyers' purchase behavior and information processing. *Computers in Human Behavior*, 84(1), 342-351.
- Deljavan Anvari, R., & Norouzi, D. (2016). The impact of e-commerce and R&D on economic development in some selected countries. *Procedia – Social and Behavioural Sciences*, 229(1), 354-362. <https://doi.org/10.1016/j.sbspro.2016.07.146>
- dos Santos, V. F., Sabino, L. R., Macedo Moraes, G., & Gonçalves, C. A. (2017). E-Commerce: A Short History Follow-up on Possible Trends. *International Journal of Business Administration (Vol. 8)*.
- EcommerceDB. (2021). E-Commerce Market Analysis. *The E-Commerce Market in Nigeria: Statista*. <https://ecommercedb.com/en/markets/ng/all>.
- Ejemeyovwi, J. O., Osabuohien, E. S., Johnson, O. D., & Bowale, E. I. K. (2019). Internet usage, innovation and human development nexus in Africa: the case of ECOWAS. *Journal of Economic Structures*, 8(1), 15. <https://doi.org/10.1186/s40008-019-0146-2>.
- Forbes Africa (2022). The Leading E-Commerce Player on an Expansion Drive. <https://cms.forbesafrica.com/wp-content/uploads/2022/02/Forbes-special-edition-KONGA-FINAL-1.pdf>.
- Gillwald, A., Mothobi, O., & Rademan, B. (2018). The State of ICT in Nigeria 2018.
- Ibam, E. O., Boyinbode, O. K., & Afolabi, M. O. (2018). e-Commerce in Africa: The case of Nigeria. *EAI Endorsed Transactions on Serious Games*, 4(15), 153536. <https://doi.org/10.4108/eai.5-1-2018.153536>
- Ibam, E. O., Boyinbode, O., & Afolabi, M. O. (2018). e-Commerce in Africa: The Case of Nigeria.
- Igwe, E., Olumuyiwa, A., & Abass, O. (2020). A Review of e-Commerce Adoption in Nigeria based on Security and Trust. *Journal of Science and Logics in ICT Research (Vol. 5)*.

International Trade Administration (2021b). Nigeria - Country Commercial Guide. <https://www.trade.gov/country-commercial-guides/nigeria-ecommerce>.

Johnson, J. (2021). Internet user penetration in Nigeria from 2016 to 2026. *Statista*. <https://www.statista.com/statistics/484918/internet-user-reach-nigeria/>.

Kemp, S. (2022). Digital 2022 - Nigeria. *Datareportal*. <https://datareportal.com/reports/digital-2022-nigeria>.

Khan, A. G. (2016). Electronic Commerce: A Study on Benefits and Challenges in an Emerging Economy. *Global Journal of Management and Business Research: Economics and Commerce* 16(1), 5.

Kianpour, K., Jusoh, A., & Asghari, M. (2013). Environmentally friendly as a new dimension of product quality. *International Journal of Quality & Reliability Management*, 19(4), pp. 442-453.

Lawal, A., & Ogbu, C. (2015). E-commerce, problem and prospects in Nigeria. *International Journal of Scientific Engineering and Applied Science (IJSEAS)*, 1(3), 1–7.

Lawal, A., & Ogbu, R. C. (2015). E-Commerce Problems and Prospects in Nigeria. *International Journal of Scientific Engineering and Applied Science (IJSEAS)* (Vol. 1).

Lawrence, J. E., & Tar, U. A. (2010). Barriers to ecommerce in developing countries. *Information, Society and Justice* (Vol. 3, pp. 23-35).

Macchion, L., Moretto, A. M., Caniato, F., Caridi, M., Danese, P., & Vinelli, A. (2017). International e-commerce for fashion products: What is the relationship with performance? *International Journal of Retail & Distribution Management*, 45(9), 1011–1031. <https://doi.org/10.1108/IJRDM-11-2015-0171>

Macchion, L., Moretto, A. M., Caniato, F., Caridi, M., Danese, P., & Vinelli, A. (2017). International E-commerce for Fashion Products: What is the Relationship with Performance? *International Journal of Retail & Distribution Management*, 45(9), pp. 1011-1031.

Mangiaracina, R., Marchet, G., Perotti, S., & Tumino, A. (2015). A review of the environmental implications of B2C e-commerce: a logistics perspective. *International Journal of Physical Distribution & Logistics Management*, 45(6), pp. 565-591.

Mourya, S. K., & Gupta, S. (2015). E-commerce. *Alpha Science International Ltd*.

NITDA, National Information Technology Development Agency (2007). National Information Technology Development Agency Act 2007. *The Federal Republic of Nigeria Official Gazette* (Vol.94).

OECD. (2011). E-commerce uptake. *OECD Science, Technology and Industry Scoreboard 2011: OECD Publishing*. https://www.oecd-ilibrary.org/docserver/sti_scoreboard-2011-64-en.pdf?expires=1652895630&id=id&accname=guest&checksum=3F3E03AA079BF5734B5210FB120B60FD.

Ogbo, A., Ugwu, C. C., Enemuo, J., & Ukpere, W. I. (2019). E-commerce as a strategy for sustainable value creation among selected traditional open market retailers in Enugu state, Nigeria. *Sustainability*, 11, 4360. <https://doi.org/10.3390/su11164360>

Okolie, U. C., & Ojomo, A. H. (2020). E-Commerce in Nigeria: Benefits and Challenges.

Okunlola, F. A., Babajide, A. A. & Isibor, A. (2020b). Empowering women through micro finance: Empirical evidence from Ibadan, Oyo state, Nigeria. *Academy of Entrepreneurship Journal* 26(1), 1–12

Olah, J., Kitukutha, N. Haddad, H., Pakurar, M., Mate, D., & Popp, J. (2018). Achieving sustainable e-commerce in environmental, social and economic dimensions by taking possible trade-offs. *Sustainability*, 11(89), 1–22. <https://doi.org/10.3390/su11010089>

Rabiu, A. M., Ibrahim, A., Dauda, M. a., Mukhtar, M. I., Abdullahi, A. M., & Abdulazeez, Y. (2019). Adoption of E-commerce in Nigeria Challenges and Future Prospects. *IEEE (pp. 1-8)*.

Sasu, D. D. (2021). Number of online shoppers in Nigeria in 2020. *Statista*. <https://www.statista.com/statistics/1139883/digital-buyers-in-nigeria/>.

Sasu, D. D. (2022b). Leading e-commerce sectors in Nigeria in 2020. *Statista*. <https://www.statista.com/statistics/1175934/leading-e-commerce-sectors-in-nigeria/>.

Schneider, G. P. (2016). Electronic commerce. *Cengage Learning (12th edition)*.

Shao, T., & Liu, Z. (2012). How to Maintain the Sustainability of an E-commerce Firm? From the Perspective of Social Network. *International Journal of Networking and Virtual Organizations*, 11(3), pp. 212-224.

Sullivan, Y. W., & Kim, D. J. (2018). Assessing the Effects of Consumers' Product Evaluations and Trust on Repurchase Intention in E-commerce Environments. *International Journal of Information Management*, 39, pp. 199-129.

The Ministry Of Communication Technology (2012). National Information Communication Technology (ICT) Policy. [https://www.researchictafrica.net/countries/nigeria/Nigeria_National_ICT_Policy_\(draft\)_2012.pdf](https://www.researchictafrica.net/countries/nigeria/Nigeria_National_ICT_Policy_(draft)_2012.pdf).

UNCTAD. (2015). Information Economy Report 2015 - Unlocking the Potential of E-commerce for Developing Countries. *United Nations Publications*. https://unctad.org/system/files/official-document/ier2015_en.pdf.

WorldData. (2022). Population growth in Nigeria. <https://www.worlddata.info/africa/nigeria/populationgrowth.php>.

Yang, Z., Shi, Y., & Yan, H. (2016). Scale, congestion, efficiency and effectiveness in e-commerce firms. *Electronic Commerce Research and Applications*, 20, pp. 171-182.

Zatonatska, T., Dluhopolskyi, O., Chyrak, I., & Kotys, N. (2019). The internet and e-commerce diffusion in European countries (modeling at the example of Austria, Poland and Ukraine). *Innovative marketing*, 15(1), 66-75. [https://doi.org/10.21511/im.15\(1\).2019.06](https://doi.org/10.21511/im.15(1).2019.06).

Zhu, Q., Sarkis, J., & Lai, K. H. (2012). Green Supply Chain Management Innovation Diffusion and its Relationship to Organizational Improvement: an Ecological Modernization Perspective. *Journal of Engineering and Technology Management*, 29(1), pp. 168-185.

Hariram, Nediyrappil & Mekha, K. & Suganthan, Vipinraj & Sudhakar, K.. (2023). Sustainalism: An Integrated Socio-Economic-Environmental Model to Address Sustainable Development and Sustainability. *Sustainability*. 15. 10682. [10.3390/su151310682](https://doi.org/10.3390/su151310682).

Purvis, Ben & Mao, Yong & Robinson, Darren. (2019). Three pillars of sustainability: in search of conceptual origins. *Sustainability Science*. 14. [10.1007/s11625-018-0627-5](https://doi.org/10.1007/s11625-018-0627-5).

Ammenberg, J. and Hjelm, O. (2013). *Miljöteknik – för en hållbar utveckling*. Lund: Studentlitteratur AB.

Arnold, F., Cardenas, I., Sörensen, K. and Dewulf, W. (2018). Simulation of B2C e-commerce Distribution in Antwerp using Cargo Bikes and Delivery Points. *European Transport Research Review*, 10(2), pp.1-13.

Bergman, B. and Klefsjö, B. (2012). *Kvalitet; från behov till användning [Quality; from need to use]*. Lund: Studentlitteratur AB.

Nisar, T.M. and Prabhakar, G. (2017). What Factors Determine e-satisfaction and Consumer Spending in e-commerce Retailing? *Journal of Retailing and Consumer Services*, 39, pp. 135-144.

Pauwels, K., Leeflang, P. S. H., Teerling, M.L., Huizingh, K. R. E. (2011). Does Online Information Drive Offline Revenues? Only for Specific Products and Consumer Segments! *Journal of Retail*, 87(1), pp. 1-17.